SkillNavora Academy · Master Freelancing

Data Analytics and Data Science

Fifteen data professions, built on the existing SkillNavora data courses

Analyst · Data scientist · Data engineer · Visualisation · Business analyst · Data mining · Power BI · Quantitative researcher · SQL developer · BI developer · Analytics engineer · Financial analyst · Python analyst · Data quality · Reporting analyst

Department 5 of 7 · Draft · Official site: www.skillnavora.in

Tool features and prices change. Official links were not checked live in this draft; confirm before publication.

Contents

1. How to use this textbook
2. Data fundamentals
3. Excel for analysis and reporting
4. SQL: from SELECT to window functions
5. Power BI: model, measures, security and publishing
6. Python and statistics for analysis
7. Advanced analytics: pipelines, warehouses and forecasting
8. Freelance projects with sample data and solutions
9. Data analyst
10. Data scientist
11. Data engineer
12. Data visualisation specialist
13. Business analyst
14. Data mining specialist
15. Microsoft Power BI specialist
16. Quantitative researcher
17. SQL developer
18. BI developer
19. Analytics engineer
20. Financial analyst
21. Python data analyst
22. Data quality specialist
23. Reporting analyst

START HERE

How to use this textbook

Data work turns records into decisions. This textbook builds the freelance and commercial layer on top of the existing SkillNavora data courses, so you do not repeat what they already teach.

Audit of existing SkillNavora data material

The Data Science and AI handbook already contains 129 lessons in 16 tracks: Advanced Excel (12), SQL (15), Power BI (6), Tableau (6), AI for data analysis (4), Python for data science (29), statistics and machine learning (28), generative AI (12), agents (6), tools (2), capstone and interviews (4), plus single lessons on related topics. This textbook does not repeat those lessons. Each profession chapter points to the relevant track and adds the freelance layer: client briefs, deliverables, pricing and portfolio work.

VERIFICATION NOTE

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PART 1 · MODULE A

Data fundamentals

The basics every data role depends on.

A1. Data types

Structured data sits in tables with fixed columns (sales records). Unstructured data is free text, images or audio. Semi-structured data (JSON) has tags but flexible shape.

A2. Cleaning, validation and transformation

Cleaning fixes missing, duplicate and wrong values. Validation checks rules (an amount is a number above zero). Transformation reshapes data for analysis (grouping, joining, calculating new columns).

A3. From data to business meaning

Every chart or table should answer a business question: what happened, why, and what to do. Write the question first.

PART 1 · MODULE B

Excel for analysis and reporting

Formulas, pivots and repeatable cleaning.

B1. Formulas every freelancer uses

Need	Formula (example)
Sum by category	`=SUMIF(B:B,"search",C:C)`
Count matches	`=COUNTIF(B:B,"social")`
Lookup (Excel 365 and recent versions)	`=XLOOKUP(A2,Customers!A:A,Customers!B:B)`
Lookup (all versions)	`=INDEX(Customers!B:B,MATCH(A2,Customers!A:A,0))`

B2. Pivot tables, conditional formatting and Power Query

A pivot table summarises rows into columns in seconds. Conditional formatting highlights values that break a rule. Power Query cleans and merges data in a repeatable way; refresh it when new data arrives.

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PART 1 · MODULE C

SQL: from SELECT to window functions

Each query is tested against a sample database.

The SQL answers below run against a small sample database of customers and orders (nine orders in three months). Each query is tested: the expected results were calculated by hand first.

C1. Total revenue by channel

SQL

```
SELECT channel, SUM(amount) AS revenue
FROM orders
GROUP BY channel
ORDER BY revenue DESC;
```

Result: search 5,600; social 3,900; email 1,600.

C2. Monthly revenue

SQL

```
SELECT substr(order_date, 1, 7) AS month, SUM(amount) AS revenue
FROM orders
GROUP BY month
ORDER BY month;
```

Result: January 3,700; February 2,100; March 5,300.

C3. Repeat customers

SQL

```
SELECT c.name, COUNT(o.id) AS orders
FROM customers c
JOIN orders o ON o.customer_id = c.id
GROUP BY c.id, c.name
HAVING COUNT(o.id) >= 2
ORDER BY c.name;
```

Result: Asha 3 orders; Meera 2; Ravi 2.

C4. Average order value by city

SQL

```
SELECT c.city, ROUND(AVG(o.amount), 2) AS avg_order_value
FROM orders o
JOIN customers c ON c.id = o.customer_id
GROUP BY c.city
ORDER BY c.city;
```

Result: Delhi 400; Mumbai 2,000; Pune 940.

C5. Running total (window function)

SQL

```
SELECT order_date, amount,
       SUM(amount) OVER (ORDER BY order_date ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS
running_total
FROM orders
WHERE customer_id = 1
ORDER BY order_date;
```

Result: 1,200 then 2,000 then 2,700 for customer 1.

C6. Query habits

- Filter early with WHERE to reduce work.
- Index columns used in joins and filters on large tables.
- Check results against a hand calculation on a small sample.

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PART 1 · MODULE D

Power BI: model, measures, security and publishing

Dashboards clients can use, built on a clean data model.

D1. Data model and relationships

Power BI works best with a star schema: one fact table (orders) linked to dimension tables (customers, dates). Relationships should be one-to-many with a single active path.

D2. Measures in DAX

Measures calculate totals dynamically. Example (verify syntax in Microsoft's DAX reference before use):

DAX

```
Total Revenue = SUM ( Orders[amount] )
Orders Count = COUNTROWS ( Orders )
Average Order Value = DIVIDE ( [Total Revenue], [Orders Count] )
Revenue Search = CALCULATE ( [Total Revenue], Orders[channel] = "search" )
```

These DAX measures were not run in this session; they follow the standard DAX pattern and must be tested in Power BI Desktop before delivery.

D3. Row-level security and publishing

Restrict each user's rows with a role and a filter expression, then test with 'View as role' in Power BI Desktop before publishing.

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PART 1 · MODULE E

Python and statistics for analysis

Pandas summaries (tested) and the statistics you must explain.

E1. Python summary (tested)

PYTHON

```
import pandas as pd
orders = pd.DataFrame({
    "channel": ["search","social","email","search","social"],
    "amount":  [1200, 800, 900, 2500, 1100],
})
summary = orders.groupby("channel")["amount"].agg(["count","sum","mean"]).round(1)
print(summary)
assert summary.loc["search","sum"] == 3700
assert summary.loc["social","count"] == 2
print("pandas summary tests passed")
```

E2. Descriptive statistics and probability

Mean, median and spread describe a typical value and how much values vary. Use the median for skewed amounts (a few large orders pull the mean up).

E3. Hypothesis testing

A test asks whether a difference could be chance. Set the question, the threshold (for example 5%), run the test, and report the result in plain words with the sample size.

VERIFICATION NOTE

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PART 1 · MODULE F

Advanced analytics: pipelines, warehouses and forecasting

The commercial side of data engineering and forecasting.

F1. ETL and pipelines

ETL means extract, transform and load: pull data from sources, clean and reshape it, and load it into a store. Schedule it, log each run and alert on failure.

F2. Warehousing and cloud analytics

A data warehouse stores cleaned data for reporting. Cloud platforms (for example BigQuery, Snowflake, Azure Synapse) offer managed storage; check current pricing and free tiers on each official page.

F3. Forecasting and predictive analytics

Forecasts use history to project the next period. Always show a range and test against a holdout period.

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PART 2 · MODULE G

Freelance projects with sample data and solutions

Seven client-style projects. The first is worked in full above; the others follow the same steps with the sample database.

#	Project	Uses
1	Sales dashboard (worked above)	SQL, dashboard
2	Customer retention analysis	Cohorts in SQL (see the analytics engineer chapter)
3	Marketing campaign analysis	Channel CAC and ROAS (see the Marketing textbook)
4	Financial performance report	Variance analysis (see the financial analyst chapter)
5	Customer segmentation	Spend segments (worked in the data mining chapter)
6	SQL reporting	Reporting view (worked in the SQL developer chapter)
7	Power BI executive dashboard	Shared measures (see Module D)

Each project needs a brief, sample data, the steps, the expected output, a rubric and a client-ready summary. Use the sample database in this book for practice, and a real or clearly fictional dataset for your portfolio.

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 1

Data analyst

Profession 1 of 10 in Development and IT.

What the profession means

A data analyst answers business questions with data: pulls the records, cleans them, analyses patterns and communicates findings to decision makers.

What professionals do every day

Writes queries, builds reports and dashboards, checks data quality, runs comparisons and presents findings with a recommendation.

Why businesses hire them, and what problems they solve

Decisions need evidence. An analyst turns records into answers that managers can act on.

- Reports that take days to build by hand.
- Numbers that disagree between teams.
- Decisions made on averages that hide the detail.

Skills you need

- SQL and spreadsheets.
- Cleaning and validation.
- Charts that answer one question.
- Clear written findings.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Microsoft Excel	Analysis and reports	Free online version	Microsoft 365 plans	www.microsoft.com/microsoft-365/excel
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Excel: basic formulas and a pivot table (Dt-B).
Intermediate	SQL: SELECT, GROUP BY, JOIN (Dt-C).

Advanced	Power BI: one dashboard (Dt-D).

Projects by level

- **Beginner:** Sales dashboard for a shop.
- **Intermediate:** Campaign comparison for a clinic.
- **Advanced:** Monthly KPI report for a firm.

Client-ready portfolio project

The sales dashboard project (Module G), with findings and a recommendation.

Worked example

Worked example. A query shows search earns the most per order; the analyst checks whether the number of orders or the order size explains it before recommending more budget.

Assignment with full solution

ASSIGNMENT

Freelance project: sales dashboard for a fictional shop

Business problem: A shop owner wants to see revenue by channel and by month, updated every week, without learning SQL.

Steps

1. Create the tables and load the sample data (see the SQL module).
2. Write the two queries: revenue by channel and monthly revenue.
3. Build a dashboard in Excel or Power BI with one chart per query.
4. Check each total against the hand calculation.
5. Write one paragraph of findings for the owner.

Expected output: A two-chart dashboard whose totals match the queries, and a findings paragraph.

SOLUTION (ANSWER KEY)

Channel totals: search 5,600; social 3,900; email 1,600 (sum 11,100). Monthly: January 3,700; February 2,100; March 5,300. Findings: search earns the most; February is the weakest month.

SQL

```
SELECT channel, SUM(amount) AS revenue
FROM orders
GROUP BY channel
ORDER BY revenue DESC;
```

SQL

```
SELECT substr(order_date, 1, 7) AS month, SUM(amount) AS revenue
FROM orders
```

```
GROUP BY month
ORDER BY month;
```

Common mistakes

- Reporting averages without the spread.
- Charts with truncated axes.
- Using numbers without checking the source.
- Recommendations with no evidence.

Freelancing opportunities

- Dashboards and monthly reports for small firms.
- Ad-hoc analysis requests on freelance marketplaces.
- Briefs for 'data analyst', 'Excel dashboard', 'SQL report'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Microsoft Learn: data analysis path
- SQLBolt (free interactive SQL lessons)

Data scientist

Profession 2 of 10 in Development and IT.

What the profession means

A data scientist builds models that predict or classify, from data preparation and experiments through to a tested model that the business can act on.

What professionals do every day

Frames the problem, prepares data, trains and compares models, evaluates them on held-out data, and explains results to non-technical staff.

Why businesses hire them, and what problems they solve

Predictions guide pricing, stock and marketing. A data scientist says how reliable each prediction is.

- Predictions that look good on training data only.
- Models nobody can explain.
- Experiments that cannot show cause.

Skills you need

- Python and pandas.
- Statistics and experiment design.
- Scikit-learn models and evaluation.
- Communication of uncertainty.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
scikit-learn	Machine learning models	Free	Free	scikit-learn.org
Kaggle	Datasets and notebooks	Free	Paid features	www.kaggle.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Clean a dataset and compute summaries.
Intermediate	Train one classifier with a test set.
Advanced	Compare two models and explain the choice.

Projects by level

- **Beginner:** Churn prediction for a subscription.
- **Intermediate:** Demand forecast for a shop.

- **Advanced:** Lead scoring for a training firm (see the AI and Automation textbook).

Client-ready portfolio project

A lead-scoring study for a fictional training firm: data preparation, a model with a held-out test score, and a plain-language limitation note.

Worked example

Worked example. A model scores 0.86 on held-out data. The report says what that means for sales, and what it cannot show.

Assignment with full solution

ASSIGNMENT

Model with a held-out score

Business problem: A training firm wants to predict who will enrol.

Steps

1. Split the data into training and test sets with a fixed seed.
2. Train one model and report accuracy on the test set.
3. Write three sentences on what the score does and does not show.

Expected output: A test score and the limitation note.

SOLUTION (ANSWER KEY)

The model is scored only on data it did not see. The note states the limit: the score is for this data and period, not a guarantee for every future lead. The lead-scoring code in the AI and Automation textbook (Chapter 4) shows the same method.

Common mistakes

- Testing on the training data.
- Leaking future information into features.
- Reporting accuracy alone when one class is rare.
- No monitoring after launch.

Freelancing opportunities

- Predictive models for small businesses.
- Forecasts and scoring projects.
- Briefs for 'machine learning', 'predictive model', 'data science'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- scikit-learn user guide
- Kaggle Learn

Data engineer

Profession 3 of 10 in Development and IT.

What the profession means

A data engineer builds the pipelines that collect, clean, store and serve data so analysts and models can rely on it.

What professionals do every day

Designs data pipelines, writes transformation code, schedules jobs, monitors failures, manages storage and documents the data.

Why businesses hire them, and what problems they solve

Analysis is only as good as its data. Engineers make sure the numbers arrive on time and mean what they say.

- Late or broken reports.
- Duplicate or inconsistent records.
- Pipelines that are impossible to maintain.

Skills you need

- SQL and Python.
- ETL and pipeline scheduling.
- Data modelling and warehousing.
- Monitoring and data quality checks.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Apache Airflow	Pipeline scheduling	Free (self-hosted)	Managed services vary	airflow.apache.org
dbt	Transformations in SQL (verify plans)	Free core	Paid cloud	www.getdbt.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Load a CSV and clean it with a repeatable script.
Intermediate	Build a scheduled job that loads one table.

Advanced	Add data quality checks and an alert.

Projects by level

- **Beginner:** Sales pipeline for a shop.
- **Intermediate:** Order-to-report pipeline for a firm.
- **Advanced:** Multi-source warehouse for a clinic.

Client-ready portfolio project

A small pipeline: load, clean, check and publish a table, with a README, tests and a failure alert.

Worked example

Worked example. A daily load rejects rows with a negative amount and logs their ids; the report uses only validated rows.

Assignment with full solution

ASSIGNMENT

Data quality check

Business problem: A shop's order file sometimes contains bad amounts.

Steps

1. Write the SQL check that finds missing or non-positive amounts.
2. Run it on the sample data and record any rows found.
3. Add the check to the daily job and send an alert when it finds rows.

Expected output: The check, the result and the alert rule.

SOLUTION (ANSWER KEY)

The sample data contains no missing amounts, so the check returns no rows. A check that returns no rows is still a passing check; the alert triggers only when rows are found.

SQL

```sql
-- data quality check: orders with a missing or non-positive amount
SELECT id, amount FROM orders WHERE amount IS NULL OR amount <= 0;
```

Common mistakes

- Pipelines without tests.
- Hard-coded paths and secrets.
- No alert on failure.
- Overwriting history without a record.

Freelancing opportunities

- Data pipeline builds for small firms.
- Maintenance contracts for data jobs.
- Briefs for 'data pipeline', 'ETL', 'Airflow', 'dbt'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Apache Airflow documentation
- dbt documentation

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 4

Data visualisation specialist

Profession 4 of 10 in Development and IT.

What the profession means

A visualisation specialist designs charts and dashboards that show patterns accurately and quickly.

What professionals do every day

Chooses chart types for the question, designs layouts, checks colour contrast, removes clutter and tests readability with the audience.

Why businesses hire them, and what problems they solve

A good chart makes the answer obvious. A bad one misleads or hides the point.

- Charts that mislead through scale or colour.
- Dashboards with too many numbers.
- Colours that fail for colour-blind readers.

Skills you need

- Chart types and when to use them.
- Colour and contrast (see the Design textbook).
- Dashboard layout.
- Accuracy: honest axes and labels.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/
Tableau Public	Charts (free public version)	Free	Paid desktop	public.tableau.com
Google Looker Studio	Dashboards	Free	Free	lookerstudio.google.com
ColorBrewer	Colour palettes for maps and charts	Free	Free	colorbrewer2.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Redraw one misleading chart honestly.
Intermediate	Pick the chart type for five questions.

Advanced	Design a one-page dashboard.

Projects by level

- **Beginner:** Dashboard for a training firm.
- **Intermediate:** Market share chart set.
- **Advanced:** Executive summary visuals.

Client-ready portfolio project

A one-page dashboard for a fictional shop with five charts, each answering one question, and a note on each design choice.

Worked example

Worked example. A bar chart with an axis starting at 90 exaggerates a 3% difference; starting the axis at zero shows it honestly.

Assignment with full solution

ASSIGNMENT

Honest chart redraw

Business problem: A chart shows growth with a truncated axis.

Steps

1. Identify the misleading choice.
2. Redraw with an axis from zero and direct labels.
3. Check the colours for contrast.
4. Write one sentence on what the chart now shows.

Expected output: The before and after chart and the sentence.

SOLUTION (ANSWER KEY)

Starting at zero shows the true size of the change. Direct labels remove the need for a legend. The contrast check keeps labels readable.

Common mistakes

- Truncated axes.
- Pie charts with many slices.
- Rainbow palettes.
- Too many charts on one page.

Freelancing opportunities

- Dashboard design for analytics clients.

- Chart cleanup for reports.
- Briefs for 'data visualisation', 'dashboard design', 'Power BI report design'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Data-to-Viz (chart choice guide)

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 5

Business analyst

Profession 5 of 10 in Development and IT.

What the profession means

A business analyst defines what a business needs from a system or change: requirements, processes, data needs and acceptance tests.

What professionals do every day

Interviews stakeholders, maps current processes, writes requirements, checks feasibility with the team and tests the delivered solution.

Why businesses hire them, and what problems they solve

Projects fail when requirements are unclear. A business analyst makes them clear before money is spent.

- Vague requests that become rework.
- Processes nobody has written down.
- Tests that do not match the need.

Skills you need

- Requirements writing (user stories, acceptance criteria).
- Process mapping.
- Stakeholder interviews.
- Basic data analysis to size the problem.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Miro	Process maps	Free tier	Paid plans	miro.com
Jira	Requirements tracking (see Development and IT)	Free tier	Paid plans	www.atlassian.com/software/jira
Excel	Sizing and comparisons	Free online version	Microsoft 365 plans	www.microsoft.com/microsoft-365/excel

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write five user stories with acceptance criteria.
Intermediate	Map the current process for one task.
Advanced	Write a requirements document for a small change.

Projects by level

- **Beginner:** Booking process for a clinic.
- **Intermediate:** Invoice approval for a firm.
- **Advanced:** Enquiry handling for a training company.

Client-ready portfolio project

A requirements pack for a fictional clinic's enquiry process: stakeholders, current and future process maps, five user stories with acceptance criteria.

Worked example

Worked example. 'As a receptionist, I see new enquiries in one list, so I reply within an hour.' The acceptance test: a new enquiry appears in the list within a minute.

Assignment with full solution

ASSIGNMENT

User stories with acceptance criteria

Business problem: A clinic wants a faster enquiry reply.

Steps

1. List the people involved in the enquiry process.
2. Write five user stories in the 'As a..., I want..., so that...' form.
3. Add acceptance criteria to each story.
4. Check each story is testable.

Expected output: Five stories with testable criteria.

SOLUTION (ANSWER KEY)

Each story names a person and a benefit. Each acceptance criterion can be checked yes or no, which is what makes the story testable.

Common mistakes

- Requirements that say 'fast' or 'easy' without a measure.
- Skipping stakeholders who will use the result.
- Changing requirements without a record.
- Writing solutions instead of needs.

Freelancing opportunities

- Requirements and process projects for small firms.
- Analysis for system implementations.
- Briefs for 'business analyst', 'requirements document', 'process mapping'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- IIBA: business analysis body of knowledge (check current resources)

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 6

Data mining specialist

Profession 6 of 10 in Development and IT.

What the profession means

A data mining specialist finds hidden patterns in large datasets: groups of customers, unusual transactions and associations between products.

What professionals do every day

Explores data, applies clustering, association and anomaly methods, validates findings and explains them to the business.

Why businesses hire them, and what problems they solve

Patterns the business does not expect can reveal new segments and risks.

- Patterns that are chance, not real.
- Results nobody can act on.
- Data that is too messy to mine.

Skills you need

- Exploratory analysis.
- Clustering and segmentation.
- Association rules and anomaly detection.
- Validation on new data.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
scikit-learn	Clustering and models	Free	Free	scikit-learn.org
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Segment customers with three rules.
Intermediate	Find unusual orders with a simple rule.
Advanced	Validate a segment on a second month.

Projects by level

- **Beginner:** Customer segmentation for a shop.
- **Intermediate:** Fraud-style anomaly check for refunds.

- **Advanced:** Product affinity analysis for a bakery.

Client-ready portfolio project

A customer segmentation for a fictional shop with three segments, their sizes and spend, and a validation note.

Worked example

Worked example. Customers with spend of 2,000 or more form the high segment; the segment is checked on the next month before the marketing team uses it.

Assignment with full solution

ASSIGNMENT

Spend segments with validation

Business problem: A shop wants three customer groups for campaigns.

Steps

1. Run the segmentation query on the sample data.
2. Count the customers and total spend in each segment.
3. Check the segments on a second period.
4. Write a note on stability.

Expected output: The segment table and the validation note.

SOLUTION (ANSWER KEY)

SQL

```
-- simple customer segment by total spend
SELECT c.name,
       SUM(o.amount) AS spend,
       CASE WHEN SUM(o.amount) >= 2000 THEN 'high'
            WHEN SUM(o.amount) >= 1000 THEN 'medium' ELSE 'low' END AS segment
FROM customers c JOIN orders o ON o.customer_id = c.id
GROUP BY c.id, c.name
ORDER BY spend DESC;
```

Results from the query: Ravi 4,500 (high); Asha 2,700 (high); Meera 2,000 (high, exactly on the threshold); Tara 1,500 (medium); Imran 400 (low). Meera sits on the 2,000 boundary, so a small change in one order moves her segment: that is the stability question to report. Re-run on another period before using the segments for campaigns.

Common mistakes

- Reporting segments without checking they are stable.
- Choosing thresholds after seeing the results.
- Ignoring privacy rules when grouping customers.
- No business owner for the segments.

Freelancing opportunities

- Segmentation for marketing teams.
- Anomaly checks for finance and operations.
- Briefs for 'customer segmentation', 'data mining', 'clustering'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- scikit-learn clustering guide

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 7

Microsoft Power BI specialist

Profession 7 of 10 in Development and IT.

What the profession means

A Power BI specialist builds data models, measures, dashboards and secure sharing for organisations using Microsoft Power BI.

What professionals do every day

Connects to data sources, shapes data in Power Query, builds the model and DAX measures, designs reports, sets row-level security and publishes to the service.

Why businesses hire them, and what problems they solve

Organisations need self-service dashboards that refresh and stay secure. A specialist builds the model that makes the reports fast and correct.

- Slow reports caused by a poor model.
- Measures that give different totals in different reports.
- Access that is too broad.

Skills you need

- Power Query.
- Data modelling and relationships.
- DAX measures and time intelligence.
- Row-level security and publishing.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/
Microsoft Learn: Power BI	Official learning	Free	Free	learn.microsoft.com/power-bi/
Microsoft Excel	Analysis and reports	Free online version	Microsoft 365 plans	www.microsoft.com/microsoft-365/excel

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Import and clean one table in Power Query.
Intermediate	Build a star schema with two dimension tables.
Advanced	Write five measures and test each total.

Projects by level

- **Beginner:** Executive dashboard for a firm.
- **Intermediate:** Sales model for a chain of shops.
- **Advanced:** Finance summary with security by region.

Client-ready portfolio project

A Power BI executive dashboard (see Module D for the measures) for a fictional firm, with a model diagram, measures, row-level security test and publishing notes.

Worked example

Worked example. Total revenue is defined once as a measure and reused in every visual, so every report shows the same total.

Assignment with full solution

ASSIGNMENT

Single definition of revenue

Business problem: Two reports show different revenue totals.

Steps

1. Find where each report calculates revenue.
2. Replace both with one shared measure.
3. Check the totals match on the same filter.
4. Write the definition in the model's description.

Expected output: The shared measure and a matching total.

SOLUTION (ANSWER KEY)

One measure, one definition. Check the totals on the same filter before and after to prove the change. The DAX pattern is in Module D; test it in Power BI Desktop.

Common mistakes

- Calculated columns used where measures are needed.
- Many-to-many relationships without a plan.
- Sharing reports without row-level security.
- No refresh schedule.

Freelancing opportunities

- Power BI builds and maintenance for small firms.
- Dashboard training for client teams.
- Briefs for 'Power BI', 'DAX', 'dashboard'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Microsoft Power BI documentation

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 8

Quantitative researcher

Profession 8 of 10 in Development and IT.

What the profession means

A quantitative researcher designs studies with numbers: surveys, experiments and statistical models that test whether a business idea or policy has a measurable effect.

What professionals do every day

Designs the study, decides the sample size, collects or cleans the data, runs the statistical tests, and writes a report with confidence and limits.

Why businesses hire them, and what problems they solve

Businesses and institutions need to know whether a change worked, not just that it looked good.

- Results that are chance.
- Samples too small to answer the question.
- Reports that overstate certainty.

Skills you need

- Study design and sampling.
- Hypothesis testing and confidence intervals.
- Regression basics.
- Clear reporting of uncertainty.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
Jamovi	Free statistics software	Free	Free	www.jamovi.org
R (open source)	Statistical language	Free	Free	www.r-project.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Compute a mean and a confidence interval.
Intermediate	Design a sample size for a survey.
Advanced	Test a difference between two groups.

Projects by level

- **Beginner:** Customer satisfaction study.
- **Intermediate:** A/B test readout.

- **Advanced:** Market sizing survey.

Client-ready portfolio project

A survey analysis for a fictional firm: sample size, a difference test between two groups and a report with confidence and limits.

Worked example

Worked example. A difference of 5 points between two groups with a wide interval is reported as 'not clear yet', not as a win.

Assignment with full solution

ASSIGNMENT

Difference between two groups

Business problem: A firm wants to know if a new onboarding improves a score.

Steps

1. State the null hypothesis: no difference.
2. Compute the difference in means and its standard error.
3. Report the result with a confidence interval.
4. Write the conclusion in plain words, including the limit.

Expected output: The estimate, interval and conclusion.

SOLUTION (ANSWER KEY)

The conclusion depends on whether the interval excludes zero, and it must say the sample size. A wide interval means the evidence is not yet clear, which is a valid answer.

Common mistakes

- Testing many hypotheses without correction.
- Small samples reported as firm results.
- Confusing correlation with cause.
- Hiding the uncertainty.

Freelancing opportunities

- Survey and experiment analysis for firms.
- Research support for start-ups.
- Briefs for 'statistics', 'survey analysis', 'hypothesis testing'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Jamovi learning resources

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 9

SQL developer

Profession 9 of 10 in Development and IT.

What the profession means

A SQL developer writes, tunes and maintains the queries and database objects that applications and reports depend on.

What professionals do every day

Writes queries and views, designs tables and keys, optimises slow queries, writes stored procedures where needed, and reviews changes.

Why businesses hire them, and what problems they solve

Slow or wrong queries slow the whole business. A SQL developer keeps the data layer fast and correct.

- Queries that scan whole tables.
- Duplicate or broken data after joins.
- Changes that break reports.

Skills you need

- SELECT, JOIN, GROUP BY and window functions (see Module C).
- Keys, constraints and indexes.
- Query plans and optimisation.
- Version control for SQL.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
PostgreSQL	Full database server	Free	Hosted plans vary	www.postgresql.org
DBeaver	Database client (free community edition)	Free	Paid plans	dbeaver.io

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write a query with a join and a grouped total.
Intermediate	Add an index and explain the effect on a large table.
Advanced	Write a view that other reports can use.

Projects by level

- **Beginner:** Reporting views for a shop.
- **Intermediate:** Customer 360 query for a firm.
- **Advanced:** Optimisation of a slow dashboard query.

Client-ready portfolio project

A reporting view with two tested queries and a note on the index choice, for a fictional shop.

Worked example

Worked example. The view computes monthly revenue once; three reports read it, so a fix in one place updates all three.

Assignment with full solution

ASSIGNMENT

Reporting view for monthly revenue

Business problem: A shop's three reports each recalculate monthly revenue differently.

Steps

1. Write one query for monthly revenue.
2. Save it as a view.
3. Point each report at the view.
4. Check the monthly totals match the earlier results.

Expected output: The view definition and a matching result.

SOLUTION (ANSWER KEY)

SQL

```
SELECT substr(order_date, 1, 7) AS month, SUM(amount) AS revenue
FROM orders
GROUP BY month
ORDER BY month;
```

The monthly results are January 3,700; February 2,100; March 5,300. Each report now reads these from the view, so the numbers match everywhere.

Common mistakes

- Using SELECT * in production.
- Joins that multiply rows silently.
- No testing of changes against known results.
- Indexes added everywhere without reason.

Freelancing opportunities

- Database and reporting work for small firms.
- Query tuning for slow systems.
- Briefs for 'SQL', 'PostgreSQL', 'database query'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- PostgreSQL documentation
- SQLBolt

BI developer

Profession 10 of 10 in Development and IT.

What the profession means

A BI developer builds the reporting layer for a business: data connections, models, dashboards and scheduled reports, usually with a BI tool such as Power BI or Tableau.

What professionals do every day

Connects sources, designs models, builds reports, sets refresh schedules and access, and supports business users with changes.

Why businesses hire them, and what problems they solve

Leaders need the same numbers in every meeting. A BI developer delivers reports that refresh and agree with each other.

- Reports that disagree.
- Refreshes that fail without anyone knowing.
- Users asking for the same change repeatedly.

Skills you need

- Data modelling for reporting.
- Power BI or Tableau.
- Refresh and access control.
- Testing reports against source data.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/
Tableau Public	Charts	Free	Paid desktop	public.tableau.com
Looker Studio	Dashboards	Free	Free	lookerstudio.google.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Build one report from a single source.
Intermediate	Set up a scheduled refresh (where the tool allows).
Advanced	Test one total against the source.

Projects by level

- **Beginner:** Monthly management pack.
- **Intermediate:** Sales dashboard for regions.
- **Advanced:** Finance report with access control.

Client-ready portfolio project

A management pack for a fictional firm: one model, four report pages, refresh notes and a reconciliation of totals to the source.

Worked example

Worked example. Before publishing, the total revenue on the dashboard is checked against the SQL total: 11,100 for the sample data.

Assignment with full solution

ASSIGNMENT

Reconcile the dashboard to the source

Business problem: A dashboard total must match the source before the leadership team uses it.

Steps

1. Compute the source total with SQL.
2. Compare it with the dashboard total.
3. Investigate any difference.
4. Record the check in the handover.

Expected output: A matching total and a check record.

SOLUTION (ANSWER KEY)

SQL

```
SELECT channel, SUM(amount) AS revenue
FROM orders
GROUP BY channel
ORDER BY revenue DESC;
```

The channel totals sum to 11,100. The dashboard total must show the same figure; a difference means a filter or a join is wrong.

Common mistakes

- Publishing without reconciliation.
- Hard-coded numbers in the dashboard.
- No owner for data refresh.
- Access given to everyone by default.

Freelancing opportunities

- BI builds for small and mid-size firms.
- Reporting maintenance contracts.
- Briefs for 'BI developer', 'Power BI', 'Tableau dashboard'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Microsoft Power BI documentation

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 11

Analytics engineer

Profession 11 of 10 in Development and IT.

What the profession means

An analytics engineer models data for analysis: transforms raw tables into clean, tested models that analysts and dashboards use, often with SQL and a transformation tool.

What professionals do every day

Writes transformation models, documents them, adds tests (for example uniqueness and not-null checks), and reviews changes with analysts.

Why businesses hire them, and what problems they solve

Analysts spend less time cleaning when the models are tested and documented once.

- Models that nobody documents.
- Silent duplicates after a join.
- Dashboards built on raw tables that change.

Skills you need

- SQL modelling.
- Testing and documentation.
- Version control.
- Understanding of business definitions.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
dbt Core	Transformations and tests (open source)	Free	Paid cloud	www.getdbt.com
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write one model that joins two tables.
Intermediate	Add a uniqueness test on the key.
Advanced	Document one model in plain words.

Projects by level

- **Beginner:** Clean model for a shop's orders.
- **Intermediate:** Customer model for a firm.
- **Advanced:** Revenue model shared by three dashboards.

Client-ready portfolio project

A tested model for a fictional shop: a clean orders table with a key test, a not-null test and a documented revenue definition.

Worked example

Worked example. An orders model joins customers once; a uniqueness test on the order id catches any join that creates duplicates.

Assignment with full solution

ASSIGNMENT

Uniqueness test on a joined model

Business problem: A join has started duplicating orders.

Steps

1. Write the model that joins orders to customers.
2. Count rows before and after the join.
3. Add a test that the order id is unique.
4. Record the result.

Expected output: The model, the row counts and the passing test.

SOLUTION (ANSWER KEY)

SQL

```
-- row counts: orders must not grow after the join
SELECT COUNT(*) FROM orders;
SELECT COUNT(*) FROM orders o JOIN customers c ON c.id = o.customer_id;
-- both return 9 for the sample data, so the join adds no duplicates
```

Common mistakes

- Trusting a join without counting rows.
- Undocumented business definitions.
- Changing models without tests.
- Naming columns differently in each model.

Freelancing opportunities

- Modelling projects for analytics teams.
- Model maintenance and tests.
- Briefs for 'dbt', 'analytics engineering', 'data modelling'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- dbt documentation

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 12

Financial analyst

Profession 12 of 10 in Development and IT.

What the profession means

A financial analyst analyses money: budgets, forecasts, variance against plan, profitability and the cash position, and explains them to managers.

What professionals do every day

Builds budget and forecast models, compares actuals with plan, analyses margins, prepares monthly reports and advises on decisions with financial impact.

Why businesses hire them, and what problems they solve

Decisions about pricing, hiring and spending need clear numbers. A financial analyst gives those numbers and shows the assumptions.

- Forecasts that are not updated.
- Variances nobody explains.
- Models with hidden assumptions.

Skills you need

- Financial statements basics.
- Budgeting and variance analysis.
- Forecast models in Excel.
- Clear assumptions and scenarios.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Microsoft Excel	Analysis and reports	Free online version	Microsoft 365 plans	www.microsoft.com/microsoft-365/excel
Google Sheets	Shared models	Free	Paid Workspace plans	sheets.google.com
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Compute variance between budget and actual.
Intermediate	Build a three-scenario forecast.
Advanced	Write a one-page commentary.

Projects by level

- **Beginner:** Monthly management accounts summary.
- **Intermediate:** Pricing margin analysis.
- **Advanced:** Cash flow forecast for a start-up.

Client-ready portfolio project

A monthly financial summary for a fictional shop: actual against budget, variances with explanations, a three-scenario forecast and a commentary.

Worked example

Worked example. Actual revenue of 11,100 against a budget of 12,000 is a variance of minus 900 (7.5%). The commentary explains the cause before any recommendation.

Assignment with full solution

ASSIGNMENT

Variance and commentary

Business problem: A shop's owner wants to know why revenue is below budget.

Steps

1. Compute the variance: actual minus budget.
2. Express it as a percentage of budget.
3. Split by channel to find the cause.
4. Write two sentences of commentary.

Expected output: The variance table and the commentary.

SOLUTION (ANSWER KEY)

11,100 minus 12,000 is minus 900, which is 7.5% of the budget. The channel split shows which part of the business drove the gap. Commentary names the cause and the next action; it does not invent a cause without evidence.

Common mistakes

- Forecasting without scenarios.
- Mixing gross and net figures.
- No record of assumptions.
- Reporting numbers without the period.

Freelancing opportunities

- Monthly management reporting for small firms.
- Forecasting for start-ups.

- Briefs for 'financial analysis', 'budget model', 'variance analysis'. Tax and legal advice belongs to a qualified professional.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Microsoft Excel help

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 13

Python data analyst

Profession 13 of 10 in Development and IT.

What the profession means

A Python data analyst uses Python and pandas to clean, analyse and visualise data, and to automate reports that would take hours by hand.

What professionals do every day

Loads files, cleans them with pandas, computes summaries, produces charts and exports results, and schedules the scripts.

Why businesses hire them, and what problems they solve

Repeated reports are slow and error-prone by hand. A script runs the same steps correctly every time.

- Manual steps that differ each month.
- Data types that break calculations.
- Scripts without tests.

Skills you need

- Python basics.
- pandas: load, filter, group, merge.
- Plotting with matplotlib or similar.
- Testing and reproducible scripts.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
Jupyter	Notebooks	Free	Free	jupyter.org
matplotlib	Charts	Free	Free	matplotlib.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write a script that loads a CSV and prints the summary.
Intermediate	Add an assert that checks a total.
Advanced	Export a chart as an image.

Projects by level

- **Beginner:** Monthly sales report automation.
- **Intermediate:** Survey data cleaning.

- **Advanced:** Customer summary script.

Client-ready portfolio project

A script that produces a monthly summary from a CSV, with asserts that check totals and an exported chart, for a fictional shop.

Worked example

Worked example. A pandas groupby gives revenue by channel; an assert checks the total equals the sum of the raw amounts before the report is written.

Assignment with full solution

ASSIGNMENT

Automated summary with checks

Business problem: A shop owner receives the same report by hand every month.

Steps

1. Load the orders file with pandas.
2. Group by channel and sum the amounts.
3. Assert that the grouped total equals the raw total.
4. Write the report to a file.

Expected output: A script, its checks and the report.

SOLUTION (ANSWER KEY)

PYTHON

```
import pandas as pd
orders = pd.DataFrame({'channel':['search','social','email','search','social'],'amount':
[1200,800,900,2500,1100]})
by_channel = orders.groupby('channel')['amount'].sum()
assert by_channel.sum() == orders['amount'].sum()
print(by_channel)
print('reconciliation passed')
```

Common mistakes

- Editing the data by hand between runs.
- No checks on totals.
- Hard-coded file paths.
- Scripts nobody else can run.

Freelancing opportunities

- Automated report scripts for small businesses.
- Data cleaning services.

- Briefs for 'Python', 'pandas', 'data automation'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- pandas documentation
- Python tutorial (official)

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 14

Data quality specialist

Profession 14 of 10 in Development and IT.

What the profession means

A data quality specialist defines and checks the rules that data must meet, measures how well it meets them, and works with owners to fix the causes.

What professionals do every day

Writes validation rules, runs checks on schedules, reports quality scores, and tracks issues to resolution.

Why businesses hire them, and what problems they solve

Bad data quietly corrupts every report built on it. Quality checks catch problems early and show where they come from.

- Problems found only by angry users.
- Rules that differ between teams.
- Fixes that treat symptoms.

Skills you need

- Validation rules: completeness, uniqueness, validity.
- SQL and Python checks.
- Root-cause analysis.
- Reporting quality scores.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
Great Expectations	Data validation framework (open source)	Free	Paid cloud	greatexpectations.io

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write three rules for an orders table.
Intermediate	Run the rules and report pass or fail.
Advanced	Trace one failure to its cause.

Projects by level

- **Beginner:** Quality checks for a shop's orders.
- **Intermediate:** Customer data audit for a firm.
- **Advanced:** Survey response validation.

Client-ready portfolio project

A quality report for a fictional shop's orders: three rules, results, a failure traced to its cause and a fix plan.

Worked example

Worked example. Rule: every order has a positive amount. The check passes on the sample data; a row with amount 0 would fail, and the report traces it to the form that allowed a zero.

Assignment with full solution

ASSIGNMENT

Three rules and a result

Business problem: A shop wants its order data checked each week.

Steps

1. Write rules: amount is positive; customer id exists; order date is present.
2. Run each rule as a SQL check.
3. Report pass or fail with the count of rows.
4. Record one fix for any failure.

Expected output: The rules, the SQL and a pass report.

SOLUTION (ANSWER KEY)

SQL

```
-- data quality check: orders with a missing or non-positive amount
SELECT id, amount FROM orders WHERE amount IS NULL OR amount <= 0;
```

The first rule returns no rows on the sample data, so it passes. The customer rule and date rule are written the same way; any row returned is a failure to fix.

Common mistakes

- Rules with no owner.
- Checking once and never again.
- Fixing rows by hand without fixing the cause.
- No quality target.

Freelancing opportunities

- Data audits and quality programmes.
- Validation for analytics teams.
- Briefs for 'data quality', 'data validation', 'data audit'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Great Expectations documentation

DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 15

Reporting analyst

Profession 15 of 10 in Development and IT.

What the profession means

A reporting analyst produces the regular reports a business depends on: weekly and monthly packs, KPI updates and exception reports, delivered on time and checked.

What professionals do every day

Maintains report templates, refreshes data, checks totals, writes commentary, and handles requests for changes.

Why businesses hire them, and what problems they solve

Leaders rely on regular reports. A reporting analyst makes them reliable, on time and easy to read.

- Late reports.
- Reports that change format each month.
- Totals that do not reconcile.

Skills you need

- Spreadsheet and BI reporting.
- Reconciliation.
- Commentary writing.
- Process discipline and checklists.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Microsoft Excel	Analysis and reports	Free online version	Microsoft 365 plans	www.microsoft.com/microsoft-365/excel
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/
Google Looker Studio	Dashboards	Free	Free	lookerstudio.google.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Build a monthly report template.
Intermediate	Reconcile totals to source.
Advanced	Write a commentary with three points.

Projects by level

- **Beginner:** Monthly KPI pack for a firm.
- **Intermediate:** Weekly sales report for a chain.
- **Advanced:** Exception report for operations.

Client-ready portfolio project

A monthly report pack for a fictional shop: template, reconciled totals, commentary and a checklist for each month.

Worked example

Worked example. The report reconciles the channel total (11,100) to the source before any commentary is written.

Assignment with full solution

ASSIGNMENT

Monthly pack with reconciliation

Business problem: A shop needs its monthly report on time with totals that match.

Steps

1. Refresh the source data.
2. Compute the channel and monthly totals.
3. Reconcile the channel total to the source total.
4. Write three points of commentary.

Expected output: The pack, the reconciliation line and the commentary.

SOLUTION (ANSWER KEY)

SQL

```
SELECT channel, SUM(amount) AS revenue
FROM orders
GROUP BY channel
ORDER BY revenue DESC;
```

Channel total 11,100 reconciles to the source total of 11,100. The commentary covers the best channel (search, 5,600), the weakest month (February, 2,100) and one action.

Common mistakes

- Changing the template every month.
- No reconciliation step.
- Commentary that restates numbers without meaning.
- Late delivery with no warning.

Freelancing opportunities

- Monthly reporting for small firms.
- Report automation for finance and sales teams.
- Briefs for 'monthly report', 'KPI report', 'reporting analyst'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Microsoft Excel help